Lesson 6: Determinants and Eigenvalues

AI/ML Mathematics

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1 Motivation and Overview

Consider a Markov chain with n states and transition matrix $P \in \mathbb{R}^{n\times n}$. A fundamental question in reinforcement learning is: how quickly does the chain converge to its stationary distribution? The answer depends on the *spectral gap* $1 - |\lambda_2|$, where λ_2 is the second-largest eigenvalue of P in absolute value. But what exactly is an eigenvalue? How do we know eigenvalues exist? And how do we compute them?

The characteristic polynomial $\det(A - \lambda I) = 0$ is the bridge connecting eigenvalues to the determinant. This single equation governs the convergence rate of iterative algorithms (value iteration, policy iteration), the stability of gradient descent (all eigenvalues of the Hessian must be positive for a local minimum), and the expressiveness of linear transformations (invertibility is equivalent to a nonzero determinant). The tools developed in this lesson will let us define determinants rigorously, establish the eigenvalue–eigenvector framework, and prove the diagonalizability criterion that underpins spectral methods throughout machine learning.

Where Determinants and Eigenvalues Appear in RL/ML

- **Spectral radius and Markov chain mixing.** The rate at which a Markov chain mixes to stationarity is governed by the spectral radius $\rho(P - \pi\mathbf{1}^\top)$; eigenvalue analysis is the primary tool for bounding this quantity.
- **Gradient descent convergence.** For quadratic objectives $f(x) = \frac{1}{2}x^\top Ax - b^\top x$, the condition number $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$ determines the convergence rate of gradient descent.
- **Cayley–Hamilton and multi-step TD.** The Cayley–Hamilton theorem shows that A^n can be expressed as a polynomial in A of degree $< n$, which is used to analyze multi-step temporal difference methods and truncated matrix powers.

Prerequisites. Vector spaces, bases, dimension, linear maps, rank–nullity theorem, matrix representation, change of basis, and similarity (Lesson 5).

Learning objectives. After completing this lesson, the student will be able to:

- **Define** the determinant via cofactor expansion and the Leibniz formula, and compute determinants of small matrices.
- **Prove** the multiplicativity of determinants ($\det(AB) = \det A \cdot \det B$) and the equivalence between invertibility and nonzero determinant.
- **Compute** eigenvalues and eigenvectors of matrices by solving the characteristic polynomial.
- **Derive** the relationship between algebraic and geometric multiplicity and apply the diagonalizability criterion.
- **State** and prove the Cayley–Hamilton theorem for matrices over $\mathbb{F}$.

2 References

- Axler [2015, Chapter 10] – Determinant theory developed after eigenvalues; provides a clean proof of the Cayley–Hamilton theorem.
- Hoffman and Kunze [1971, Chapter 5] – Classical treatment of determinants via multilinear alternating forms.
- Strang [2016, Chapter 5] – Computational perspective on determinants with cofactor expansion and applications.

- Horn and Johnson [2013, Chapter 1] – Eigenvalue theory, characteristic polynomial, and matrix factorizations.
- Lax [2007, Chapter 6] – Spectral theory with connections to operator theory and applications.

3 Determinants

Throughout this lesson, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$, and all matrices are square unless stated otherwise.

3.1 Permutations and the Sign Function

Definition 3.1 (Permutation and Sign)**.** A *permutation* of $\{1, \ldots, n\}$ is a bijection $\sigma\colon \{1, \ldots, n\} \to \{1, \ldots, n\}$. The set of all such permutations is denoted S_n. A *transposition* is a permutation that swaps exactly two elements. Every permutation can be written as a composition of transpositions. The *sign* (or *signature*) of σ is

$$\operatorname{sgn}(\sigma) = (-1)^k,$$

where k is the number of transpositions in any decomposition of σ. A permutation is *even* if $\operatorname{sgn}(\sigma) = +1$ and *odd* if $\operatorname{sgn}(\sigma) = -1$.

Remark 3.2. The sign is well-defined: although a permutation can be written as a product of transpositions in many ways, the parity of the number of transpositions is invariant. One proof uses the identity $\operatorname{sgn}(\sigma) = \prod_{i<j} \frac{\sigma(j)-\sigma(i)}{j-i}$, which is manifestly well-defined [Hoffman and Kunze, 1971, Chapter 5].

Example 3.3 (Permutations of $\{1, 2, 3\}$)**.** The group S_3 has $3! = 6$ elements. The identity $e = (1)(2)(3)$ has $\operatorname{sgn}(e) = +1$. The transposition $\sigma = (1\ 2)$ (swapping 1 and 2) has $\operatorname{sgn}(\sigma) = -1$. The cyclic permutation $\tau = (1\ 2\ 3)$ (mapping $1 \mapsto 2 \mapsto 3 \mapsto 1$) decomposes as $(1\ 3)(1\ 2)$, so $\operatorname{sgn}(\tau) = +1$.

3.2 The Leibniz Formula

Definition 3.4 (Determinant – Leibniz Formula)**.** Let $A = (a_{ij}) \in \mathbb{F}^{n\times n}$. The *determinant* of A is

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Example 3.5 (Determinants of 2×2 and 3×3 Matrices)**.** For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the two permutations of $\{1, 2\}$ give $\det A = ad - bc$.

For a 3×3 matrix $A = (a_{ij})$, expanding over all six permutations in S_3 yields

$$\begin{aligned} \det A = {} & a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ & - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}. \end{aligned}$$

3.3 Cofactor Expansion

Definition 3.6 (Minor and Cofactor)**.** Let $A \in \mathbb{F}^{n\times n}$. The (i, j)-*minor* M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A. The (i, j)-*cofactor* is $C_{ij} = (-1)^{i+j} M_{ij}$.

Theorem 3.7 (Cofactor Expansion). *For any fixed row i or column j,*

$$\det A = \sum_{j=1}^{n} a_{ij}\, C_{ij} \quad \textit{(expansion along row } i\textit{)},$$

$$\det A = \sum_{i=1}^{n} a_{ij}\, C_{ij} \quad \textit{(expansion along column } j\textit{)}.$$

Proof. We prove the row expansion. In the Leibniz formula, fix row i and group terms by the column index $\sigma(i) = j$. For each j, the contribution from permutations with $\sigma(i) = j$ is $a_{ij} \cdot (-1)^{i+j} \det A^{(ij)}$, where $A^{(ij)}$ is the matrix with row i and column j deleted. To verify the sign, note that moving the factor $a_{i,\sigma(i)}$ from position i in the product to the front requires reindexing the remaining permutation on $\{1, \ldots, n\} \setminus \{i\}$ to columns $\{1, \ldots, n\} \setminus \{j\}$, which introduces a sign factor of $(-1)^{i+j}$. Summing over $j = 1, \ldots, n$ recovers $\det A = \sum_j a_{ij} C_{ij}$. The column expansion follows from the transpose property (Proposition 3.9). □

Example 3.8 (Cofactor Expansion of a 3×3 Matrix). Let $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}$. Expanding along row 1:

$$\begin{aligned} \det A &= 2 \det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} - 1 \det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} + 3 \det \begin{pmatrix} 0 & -1 \\ 5 & 2 \end{pmatrix} \\ &= 2(-1-8) - 1(0-20) + 3(0+5) = -18 + 20 + 15 = 17. \end{aligned}$$

3.4 Properties of Determinants

Proposition 3.9 (Transpose). *For any $A \in \mathbb{F}^{n \times n}$, $\det(A^\top) = \det A$.*

Proof. In the Leibniz formula, $\det(A^\top) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{\sigma(i),i}$. Since $\sigma \mapsto \sigma^{-1}$ is a bijection on S_n with $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$, reindexing with $\tau = \sigma^{-1}$ gives $\sum_{\tau \in S_n} \operatorname{sgn}(\tau) \prod_{i=1}^{n} a_{i,\tau(i)} = \det A$. □

Proposition 3.10 (Determinant and Row Operations). *Let $A \in \mathbb{F}^{n \times n}$.*

1. ***Multilinearity:*** *The determinant is linear in each row (holding the other rows fixed). That is, if row i of A is $\alpha r + \beta r'$ and B, C are obtained by replacing row i with r and r' respectively, then $\det A = \alpha \det B + \beta \det C$.*
2. ***Alternating:*** *If two rows of A are interchanged, the determinant changes sign. In particular, if two rows are equal, $\det A = 0$.*
3. ***Row addition:*** *Adding a scalar multiple of one row to another does not change the determinant.*
4. ***Row scaling:*** *Multiplying a single row by $\alpha \in \mathbb{F}$ multiplies the determinant by α.*

Proof. (1) follows directly from the Leibniz formula: each term $\operatorname{sgn}(\sigma) \prod_i a_{i,\sigma(i)}$ is linear in the entries of any fixed row.

(2) Let A' be obtained from A by swapping rows i and j. This is equivalent to composing each permutation σ with the transposition $(i\ j)$, which changes the sign. If rows i and j are equal, then $A = A'$ and $\det A = -\det A$, so $\det A = 0$.

(3) Adding α times row j to row i gives, by multilinearity, $\det A' = \det A + \alpha \det B$, where B has row j appearing in both positions i and j. By (2), $\det B = 0$.

(4) is immediate from (1) with $r' = 0$. □

Example 3.11 (Determinant via Row Reduction)**.** Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 7 & 10 \end{pmatrix}$. Subtract 2 times row 1 from row 2, and 3 times row 1 from row 3: $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. Two rows are now equal, so $\det A = 0$. By the criterion we will prove shortly, A is not invertible.

3.5 Multiplicativity of the Determinant

Theorem 3.12 (Multiplicativity)**.** *For $A, B \in \mathbb{F}^{n\times n}$,*

$$\det(AB) = \det A \cdot \det B.$$

Proof. **Case 1:** A is not invertible. Then $\operatorname{rank}(A) < n$, so $\operatorname{rank}(AB) \leq \operatorname{rank}(A) < n$. Hence AB is not invertible, and we will show in Theorem 3.13 that $\det(AB) = 0 = \det A \cdot \det B$.

Case 2: A is invertible. Every invertible matrix is a product of elementary matrices (each representing a single row operation). An elementary matrix E satisfies $\det(EB) = (\det E)(\det B)$ by Proposition 3.10: swapping two rows multiplies the determinant by $-1 = \det E$; adding a multiple of one row to another multiplies by $1 = \det E$; scaling a row by α multiplies by $\alpha = \det E$. Writing $A = E_1 \cdots E_k$, repeated application gives

$$\det(AB) = \det(E_1 \cdots E_k B) = (\det E_1) \cdots (\det E_k)(\det B) = \det(E_1 \cdots E_k) \det B = \det A \cdot \det B.$$

□

3.6 Determinants and Invertibility

Determinant–Invertibility Equivalence

Theorem 3.13 (Determinant and Invertibility)**.** *A matrix $A \in \mathbb{F}^{n\times n}$ is invertible if and only if $\det A \neq 0$. When A is invertible, $\det(A^{-1}) = (\det A)^{-1}$.*

Proof. ($\Rightarrow$) If A is invertible, then $AA^{-1} = I$, so by multiplicativity $\det A \cdot \det(A^{-1}) = \det I = 1$. In particular, $\det A \neq 0$ and $\det(A^{-1}) = (\det A)^{-1}$.

($\Leftarrow$) If A is not invertible, then $\operatorname{rank}(A) < n$, so the rows of A are linearly dependent. Row reduction (which preserves $\det = 0$ or $\det \neq 0$) produces at least one zero row. A matrix with a zero row has determinant 0 (every term in the Leibniz formula contains a factor from this row, which is 0). Hence $\det A = 0$. □

Example 3.14 (Invertibility Check)**.** For $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$, we have $\det A = 6 - 6 = 0$, so A is singular. Indeed, row 2 is 3 times row 1. For $B = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$, we have $\det B = 7 - 6 = 1 \neq 0$, so B is invertible.

The determinant provides a scalar invariant that detects invertibility. We now turn to the richer structure revealed by the eigenvalue problem.

4 Eigenvalues and Eigenvectors

4.1 Basic Definitions

Definition 4.1 (Eigenvalue and Eigenvector)**.** Let $A \in \mathbb{F}^{n\times n}$. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of A if there exists a nonzero vector $v \in \mathbb{F}^n$ such that $Av = \lambda v$. Such a vector v is called an *eigenvector* of A corresponding to λ.

Definition 4.2 (Eigenspace)**.** The *eigenspace* of A corresponding to eigenvalue λ is

$$E_\lambda = \text{null}(A - \lambda I) = \{v \in \mathbb{F}^n : Av = \lambda v\}.$$

This is a subspace of $\mathbb{F}^n$ (it is the null space of the linear map $A - \lambda I$). The eigenspace includes the zero vector, but by convention eigenvectors are nonzero.

Example 4.3 (Eigenvalues of a 2×2 Matrix)**.** Let $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. If $Av = \lambda v$ for nonzero v, then $(A - \lambda I)v = 0$ has a nontrivial solution, which requires $\det(A - \lambda I) = 0$. We compute

$$\det(A - \lambda I) = \det \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 5$. For $\lambda_1 = 2$: $(A - 2I)v = 0$ gives $2v_1 + v_2 = 0$, so $E_2 = \text{span}\{(1, -2)\}$. For $\lambda_2 = 5$: $-v_1 + v_2 = 0$, so $E_5 = \text{span}\{(1, 1)\}$.

4.2 The Characteristic Polynomial

Definition 4.4 (Characteristic Polynomial)**.** The *characteristic polynomial* of $A \in \mathbb{F}^{n\times n}$ is

$$p_A(\lambda) = \det(A - \lambda I).$$

This is a polynomial of degree n in λ with leading coefficient $(-1)^n$. The eigenvalues of A are precisely the roots of p_A.

Remark 4.5. Similar matrices have the same characteristic polynomial: if $B = P^{-1}AP$, then $\det(B - \lambda I) = \det(P^{-1}(A - \lambda I)P) = \det(P^{-1})\det(A - \lambda I)\det(P) = \det(A - \lambda I)$. Hence eigenvalues are invariants of the linear operator, not of a particular matrix representation.

Proposition 4.6 (Trace and Determinant from Eigenvalues)**.** *If $A \in \mathbb{F}^{n\times n}$ has eigenvalues $\lambda_1, \ldots, \lambda_n$ (counted with algebraic multiplicity, over the algebraic closure of $\mathbb{F}$), then*

$$\text{tr}(A) = \sum_{i=1}^{n} \lambda_i, \qquad \det(A) = \prod_{i=1}^{n} \lambda_i.$$

Proof. The characteristic polynomial factors (over the algebraic closure) as $p_A(\lambda) = (-1)^n(\lambda - \lambda_1)\cdots(\lambda - \lambda_n)$. Setting $\lambda = 0$: $\det A = p_A(0) = (-1)^n(-\lambda_1)\cdots(-\lambda_n) = \prod_i \lambda_i$. Comparing the coefficient of λ^{n-1}: in $\det(A - \lambda I)$, the λ^{n-1} term arises only from the diagonal product $(a_{11} - \lambda)\cdots(a_{nn} - \lambda)$, giving coefficient $(-1)^{n-1}(a_{11} + \cdots + a_{nn}) = (-1)^{n-1}\,\text{tr}(A)$. From the factored form, the λ^{n-1} coefficient is $(-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)$. Equating gives $\text{tr}(A) = \sum_i \lambda_i$. □

Spectral Radius and Markov Chain Mixing

For a stochastic matrix P (rows sum to 1, nonneg entries), $\lambda_1 = 1$ is always an eigenvalue. The *spectral radius* of P restricted to the complement of the stationary distribution is $\rho = \max_{i\geq 2} |\lambda_i|$. The mixing time satisfies $t_{\text{mix}} = \Theta(1/(1 - \rho))$. In policy evaluation for RL, faster mixing means that Monte Carlo rollouts more quickly yield representative samples, reducing variance.

4.3 Algebraic and Geometric Multiplicity

Definition 4.7 (Algebraic and Geometric Multiplicity)**.** Let λ be an eigenvalue of A.

- The *algebraic multiplicity* $\text{am}(\lambda)$ is the multiplicity of λ as a root of the characteristic polynomial p_A.

- The *geometric multiplicity* $\mathrm{gm}(\lambda)$ is the dimension of the eigenspace $E_\lambda = \mathrm{null}(A - \lambda I)$.

Theorem 4.8 (Multiplicity Inequality)**.** *For every eigenvalue λ of $A \in \mathbb{F}^{n\times n}$,*

$$1 \le \mathrm{gm}(\lambda) \le \mathrm{am}(\lambda).$$

Proof. The lower bound $\mathrm{gm}(\lambda) \ge 1$ holds because the eigenspace contains at least one eigenvector (by definition of eigenvalue).

For the upper bound, let $k = \mathrm{gm}(\lambda)$ and let $\{v_1, \dots, v_k\}$ be a basis of E_λ. Extend this to a basis $\{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ of $\mathbb{F}^n$. Let P be the invertible matrix with columns $v_1, \dots, v_n$. Then

$$P^{-1}AP = \begin{pmatrix} \lambda I_k & * \\ 0 & C \end{pmatrix},$$

because $Av_i = \lambda v_i$ for $i = 1, \dots, k$. The characteristic polynomial is

$$p_A(t) = \det(P^{-1}AP - tI) = (\lambda - t)^k \det(C - tI).$$

Hence $(\lambda - t)^k$ divides $p_A(t)$, so $\mathrm{am}(\lambda) \ge k = \mathrm{gm}(\lambda)$. □

Example 4.9 (Strict Inequality of Multiplicities)**.** Consider $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. The characteristic polynomial is $(2-\lambda)^2$, so $\lambda = 2$ has $\mathrm{am}(2) = 2$. But $A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, which has nullity $= 1$, so $\mathrm{gm}(2) = 1 < 2 = \mathrm{am}(2)$.

4.4 Independence of Eigenvectors for Distinct Eigenvalues

Theorem 4.10 (Eigenvectors for Distinct Eigenvalues Are Independent)**.** *Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of $A \in \mathbb{F}^{n\times n}$ with corresponding eigenvectors $v_1, \dots, v_k$. Then $v_1, \dots, v_k$ are linearly independent.*

Proof. We proceed by induction on k. The base case $k = 1$ is immediate: a single eigenvector is nonzero, hence independent.

For the inductive step, suppose the result holds for $k - 1$ distinct eigenvalues. Assume

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0. \tag{1}$$

Apply A to both sides:

$$c_1\lambda_1 v_1 + c_2\lambda_2 v_2 + \cdots + c_k\lambda_k v_k = 0. \tag{2}$$

Multiply (1) by λ_k and subtract from (2):

$$c_1(\lambda_1 - \lambda_k)v_1 + c_2(\lambda_2 - \lambda_k)v_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} = 0.$$

By the inductive hypothesis, $v_1, \dots, v_{k-1}$ are independent. Since all $\lambda_i - \lambda_k \ne 0$ for $i < k$, we conclude $c_i = 0$ for $i = 1, \dots, k-1$. Substituting back into (1) gives $c_k v_k = 0$, and since $v_k \ne 0$, $c_k = 0$. □

Eigenvalues and Gradient Descent Convergence

For a quadratic objective $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ with A symmetric positive definite, gradient descent with step size α has update $x_{k+1} = (I - \alpha A)x_k + \alpha b$. The error $e_k = x_k - x^*$ satisfies $e_k = (I - \alpha A)^k e_0$. Since the eigenvectors of A (which are real and independent by Theorem 4.10 when eigenvalues are distinct) form a basis, each component of e_k in this eigenbasis decays as $(1 - \alpha\lambda_i)^k$. Convergence requires $|1 - \alpha\lambda_i| < 1$ for all i,

optimized by $\alpha^* = 2/(\lambda_{\min} + \lambda_{\max})$ with rate $(\kappa - 1)/(\kappa + 1)$ where $\kappa = \lambda_{\max}/\lambda_{\min}$.

Having established when eigenvalues exist and that eigenvectors for distinct eigenvalues are independent, we now ask: when can we find a basis of eigenvectors for the entire space?

5 Diagonalizability

Definition 5.1 (Diagonalizable Matrix)**.** A matrix $A \in \mathbb{F}^{n\times n}$ is *diagonalizable* if there exists an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that $P^{-1}AP$ is diagonal. Equivalently, A is diagonalizable if and only if $\mathbb{F}^n$ has a basis consisting of eigenvectors of A.

Diagonalizability Criterion

Theorem 5.2 (Diagonalizability Criterion)**.** *A matrix $A \in \mathbb{F}^{n\times n}$ is diagonalizable if and only if both of the following conditions hold:*

1. *The characteristic polynomial p_A splits completely over $\mathbb{F}$ (i.e., all eigenvalues lie in $\mathbb{F}$).*
2. *For every eigenvalue λ, $\mathrm{gm}(\lambda) = \mathrm{am}(\lambda)$.*

Proof. ($\Rightarrow$) Suppose A is diagonalizable: $P^{-1}AP = D$ where $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_i \in \mathbb{F}$. Then $p_A(\lambda) = p_D(\lambda) = \prod_{i=1}^n (\lambda_i - \lambda)$, which splits over $\mathbb{F}$. If μ appears m times on the diagonal of D, then $\mathrm{am}(\mu) = m$. The columns of P corresponding to diagonal entries equal to μ are m linearly independent eigenvectors for μ, so $\mathrm{gm}(\mu) \geq m = \mathrm{am}(\mu)$. Combined with Theorem 4.8, $\mathrm{gm}(\mu) = \mathrm{am}(\mu)$.

($\Leftarrow$) Let $\lambda_1, \ldots, \lambda_r$ be the distinct eigenvalues of A with algebraic multiplicities $m_1, \ldots, m_r$ summing to n. By assumption, each eigenspace E_{λ_i} has dimension m_i. We claim the union of bases of all eigenspaces forms a basis of $\mathbb{F}^n$.

The total count is $m_1 + \cdots + m_r = n$, so it suffices to show linear independence. Suppose $\sum_{i=1}^r w_i = 0$ where $w_i \in E_{\lambda_i}$. Write each w_i as a combination of a basis of E_{λ_i}. The vectors from different eigenspaces lie in distinct eigenspaces; by an extension of the argument in Theorem 4.10 (eigenvectors from distinct eigenspaces are independent), each $w_i = 0$. Since the basis vectors within each E_{λ_i} are independent, all coefficients vanish. Hence these n eigenvectors form a basis of $\mathbb{F}^n$, and A is diagonalizable. □

Corollary 5.3. *If $A \in \mathbb{F}^{n\times n}$ has n distinct eigenvalues in $\mathbb{F}$, then A is diagonalizable.*

Proof. Each eigenvalue has algebraic multiplicity 1, and geometric multiplicity is at least 1 by Theorem 4.8. Hence $\mathrm{gm}(\lambda_i) = 1 = \mathrm{am}(\lambda_i)$ for each i, and the criterion is satisfied. □

Not Every Matrix Is Diagonalizable

The matrix $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has characteristic polynomial λ^2, so $\lambda = 0$ is the only eigenvalue with $\mathrm{am}(0) = 2$. But $\mathrm{rank}(A - 0\cdot I) = 1$, so $\mathrm{gm}(0) = 1 < 2$. Hence A is not diagonalizable. Over $\mathbb{R}$, even 2×2 matrices may fail: the rotation matrix $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 + 1$, which has no real roots. Over $\mathbb{C}$, the eigenvalues are $\pm i$ (distinct), so R is diagonalizable over $\mathbb{C}$ but not over $\mathbb{R}$.

Example 5.4 (Diagonalization Procedure)**.** Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. The characteristic polynomial is $(1 - \lambda)(3 - \lambda)$, giving distinct eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$.

For $\lambda_1 = 1$: $(A - I)v = 0$ gives $\begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} v = 0$, so $v_1 = (1, 0)^\top$.

For $\lambda_2 = 3$: $(A - 3I)v = 0$ gives $\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} v = 0$, so $v_2 = (1, 1)^\top$.

Setting $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, we get $P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$.

With diagonalizability understood, we conclude with the Cayley–Hamilton theorem, which provides a universal polynomial identity satisfied by every square matrix.

6 The Cayley–Hamilton Theorem

Theorem 6.1 (Cayley–Hamilton)**.** *Let $A \in \mathbb{F}^{n\times n}$ and let $p_A(\lambda) = \det(A - \lambda I)$ be its characteristic polynomial. Then $p_A(A) = 0$ (the zero matrix). That is, if $p_A(\lambda) = c_0 + c_1\lambda + \cdots + c_{n-1}\lambda^{n-1} + (-1)^n\lambda^n$, then*

$$c_0 I + c_1 A + \cdots + c_{n-1}A^{n-1} + (-1)^n A^n = 0.$$

Proof. Let $\operatorname{adj}(A-\lambda I) = B_0+B_1\lambda+\cdots+B_{n-1}\lambda^{n-1}$ be the adjugate, whose entries are cofactors (polynomials of degree $\leq n-1$ in λ). The cofactor-adjugate identity gives

$$(A - \lambda I)(B_0 + B_1\lambda + \cdots + B_{n-1}\lambda^{n-1}) = p_A(\lambda) \cdot I.$$

Write $p_A(\lambda) = c_0 + c_1\lambda + \cdots + (-1)^n\lambda^n$. Equating coefficients of λ^k for $k = 0, \ldots, n$ yields $n+1$ matrix equations. Multiplying the k-th equation on the left by A^k and summing, the left side telescopes (each $A^k B_{k-1}$ appears once with $+1$ and once with -1), giving

$$0 = c_0 I + c_1 A + c_2 A^2 + \cdots + (-1)^n A^n = p_A(A).$$ □

Example 6.2 (Cayley–Hamilton for 2×2)**.** Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$. The characteristic polynomial is $p_A(\lambda) = \lambda^2 - 5\lambda - 2$. The theorem asserts $A^2 - 5A - 2I = 0$. We verify:

$$A^2 = \begin{pmatrix} 7 & 15 \\ 10 & 22 \end{pmatrix}, \quad 5A = \begin{pmatrix} 5 & 15 \\ 10 & 20 \end{pmatrix}, \quad A^2 - 5A - 2I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad \checkmark$$

Example 6.3 (Expressing A^{-1} via Cayley–Hamilton)**.** If A is invertible and $p_A(\lambda) = c_0 + c_1\lambda + \cdots + (-1)^n\lambda^n$ with $c_0 = \det A \neq 0$, then $p_A(A) = 0$ gives

$$c_0 I + c_1 A + \cdots + (-1)^n A^n = 0.$$

Multiplying by A^{-1} and solving:

$$A^{-1} = -\frac{1}{c_0}\big(c_1 I + c_2 A + \cdots + (-1)^n A^{n-1}\big).$$

This expresses A^{-1} as a polynomial in A of degree $n - 1$.

Cayley–Hamilton in Multi-Step TD

In temporal difference learning with eligibility traces, the λ-return involves the matrix geometric series $\sum_{k=0}^{\infty}(\gamma P^\pi)^k$ where P^π is the transition matrix. By the Cayley–Hamilton theorem, A^n (and all higher powers) can be expressed as linear combinations of $I, A, \ldots, A^{n-1}$. This means the effective "memory" of the matrix powers is at most n steps, and in practice, truncated sums at horizon n capture the full behavior. This

observation justifies efficient implementations of multi-step TD methods in finite-state MDPs.

7 Exercises

Exercise 7.1 (Determinant Computation). $(*)$
Compute the determinant of each matrix:

(a) $A = \begin{pmatrix} 3 & 1 & -2 \\ 0 & 4 & 5 \\ 1 & 0 & 3 \end{pmatrix}$.

(b) $B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

Exercise 7.2 (Eigenvalue Analysis). $(*)$
Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Find all eigenvalues, their algebraic and geometric multiplicities, and determine whether A is diagonalizable.

Exercise 7.3 (Determinant Identities). $(**)$
Let $A, B \in \mathbb{F}^{n\times n}$ with A invertible.

(a) Prove that $\det(A^{-1}BA) = \det B$.
(b) Prove that A and $A^\top$ have the same eigenvalues.
(c) Show by example that A and $A^\top$ can have different eigenvectors.

Exercise 7.4 (Markov Chain Spectral Analysis). $(**)$
A Markov chain on states $\{1, 2, 3\}$ has transition matrix $P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \\ 0.3 & 0.2 & 0.5 \end{pmatrix}$.

(a) Verify that $\lambda_1 = 1$ is an eigenvalue and find a corresponding left eigenvector (the stationary distribution).
(b) Compute all eigenvalues of P and determine the spectral gap $1 - |\lambda_2|$.
(c) Use the spectral gap to bound the number of steps needed for the distribution to be within $\epsilon = 0.01$ of stationarity.

Hint: Row sums equal 1, so $(1, 1, 1)^\top$ is always an eigenvector for $\lambda = 1$.

Exercise 7.5 (Cayley–Hamilton Application). $(**)$
Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

(a) Compute the characteristic polynomial $p_A(\lambda)$.
(b) Verify $p_A(A) = 0$ by direct computation.
(c) Use the Cayley–Hamilton theorem to express A^{-1} as $\alpha I + \beta A$ for explicit α, β.
(d) Use the result to compute A^5 as $\gamma I + \delta A$ for explicit γ, δ.

Exercise 7.6 (Diagonalizability Proof). $(***)$
Let $A, B \in \mathbb{F}^{n\times n}$ be diagonalizable. Prove or disprove:

(a) $A + B$ is diagonalizable.
(b) AB is diagonalizable.
(c) If A and B are *simultaneously* diagonalizable (i.e., there exists a single invertible P such that both $P^{-1}AP$ and $P^{-1}BP$ are diagonal), then $AB = BA$. Prove this and state the converse.

Hint: For (a) and (b), consider 2×2 counterexamples. For (c), use the fact that diagonal matrices commute.

8 Summary

- The *determinant* $\det A$ is defined by the Leibniz formula $\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_i a_{i,\sigma(i)}$ and can be computed recursively via cofactor expansion along any row or column.
- The determinant is *multilinear* and *alternating* in the rows; it is invariant under row addition and changes sign under row swaps.
- $\det(AB) = \det A \cdot \det B$ (multiplicativity), and A is invertible if and only if $\det A \neq 0$.
- An *eigenvalue* λ of A satisfies $Av = \lambda v$ for some nonzero v; equivalently, λ is a root of the *characteristic polynomial* $p_A(\lambda) = \det(A - \lambda I)$.
- The *trace* equals the sum of eigenvalues and the *determinant* equals the product of eigenvalues.
- For each eigenvalue, $1 \leq \operatorname{gm}(\lambda) \leq \operatorname{am}(\lambda)$. Eigenvectors for distinct eigenvalues are linearly independent.
- A is *diagonalizable* if and only if p_A splits over $\mathbb{F}$ and $\operatorname{gm}(\lambda) = \operatorname{am}(\lambda)$ for every eigenvalue λ. A sufficient condition is that A has n distinct eigenvalues.
- The *Cayley–Hamilton theorem* states $p_A(A) = 0$: every matrix satisfies its own characteristic polynomial. This allows expressing high powers A^k (for $k \geq n$) as polynomials in A of degree at most $n - 1$.

9 Further Reading

- **Lesson 7** (Inner Products and Spectral Theory) introduces inner product structure and proves the spectral theorem for real symmetric matrices: every such matrix is orthogonally diagonalizable. This strengthens the diagonalizability criterion by providing an orthonormal eigenbasis.
- **Lesson 8** (SVD and Applications) combines eigenvalue theory with inner products to derive the singular value decomposition, the most important matrix factorization in data science and ML.
- The Jordan normal form (not covered here) provides a canonical form for non-diagonalizable matrices. See Hoffman and Kunze [1971, Chapter 8] or Horn and Johnson [2013, Chapter 3].
- Horn and Johnson [2013, Chapter 1] gives a thorough treatment of eigenvalue inequalities (Weyl, Cauchy interlacing) used in matrix perturbation theory, which underlies the analysis of approximate methods in RL.
- For the historical development of determinants and eigenvalue theory, see Lax [2007], which traces the ideas from Cayley and Sylvester through modern operator theory.

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